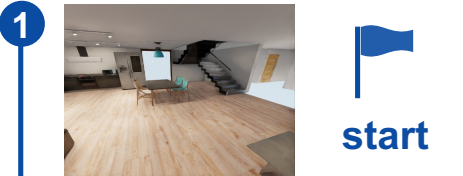


(a) Ego & Overlays



(b) Trace Log

Initial state · 0 s

get_robot_position → [4.95, 5.04, 0.01], yaw = -1.93 rad

task: pick up red radio · radio off-camera from start pose

Perception sweep · 0 – 153 s

find_object_base_rotate("red radio") · 10 base rotations →
found radio @ [3.02, 4.91, 0.30]

d = 1.91 m from robot — need to move closer

Trigger cuRobo collision-avoidance · 153 – 545 s

3 close goals ([3.64, 4.79] · [3.52, 4.89] · [3.62, 4.88])

All inside the table's collision-avoidance buffer. cuRobo returns
PLANNING_ERROR internally

Multi-angle finds open region · 545 – 607 s

★ for θ in [180°, -90°, 90°]:
target = radio + 0.7 · [cos θ , sin θ]

★ ang_180 → [2.32, 4.91] (0.7 m west of radio, far side)
cuRobo finds clean path. robot moves 1.45 m in one jump → [2.80,
4.90], d = 0.22 m

Re-perceive + single-arm grasp · 607 – 1205 s

sample_grasp_pose(red radio) → 2 candidates

grasp_object(arm=0, pose 1) → True ·

lift_arm → in_hand=True · reward = 1.0 · task_completed = 1

(c) ★ Multi-Angle Approach

fix_code.py

```
for angle_deg in [180, -90, 90, -45, 45]: #  
    south, sides only  
    angle = np.radians(angle_deg)  
    tx = radio_pos[0] + 0.7 * np.cos(angle)  
    ty = radio_pos[1] + 0.7 * np.sin(angle)  
    face_yaw = np.arctan2(radio_pos[1]-ty,  
        radio_pos[0]-tx)  
    moved = safe_navigate([tx, ty, face_yaw],  
        f"ang_{angle_deg}")  
    if moved and dist_to(radio_pos[:2]) <  
        0.8:  
        break  
    ...
```



New Skill

If cuRobo returns PLANNING_ERROR for nearby goals (inside the collision-avoidance buffer), try rotating the approach vector by 45°, 90°, or 180°.

Each angle puts the goal on a different side of the object.

When one side is blocked, another is often open.